# INDUSTRIAL AUTOMATED IRRIGATION SYSTEM FOR AGRICULTURAL PRODUCTION

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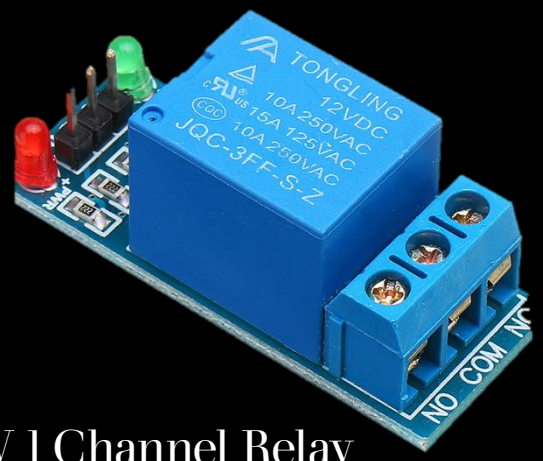
Water is one of the most important and most endangered resources on our planet. Daily conservation and smart use are key to the future of all of us. Technology can help us — with accessible solutions like Arduino, we can create systems for automatic and efficient water management. These smart systems not only make everyday life easier, but also actively protect the environment and preserve water resources for future generations.

This project represents an industrial automated irrigation system for agricultural production, developed using Arduino Uno. The goal is to minimize the loss of water resources and agricultural crops caused by drought, as well as to reduce dependence on the human factor in processes that can be automated. This enables more efficient, sustainable, and reliable irrigation management in large plantations.

# Required Materials

Arduino Uno

Soil Moisture Hygrometer Detection Module



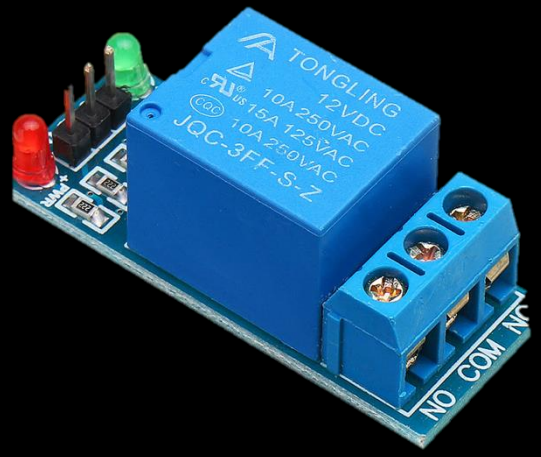

5V 1 Channel Relay

DC 3-6V Submersible Mini Water Pump

The system is scalable — more sensors or additional pumps can be added for larger areas.

A central microcontroller board that manages all connected components.
It supports digital and analog inputs and outputs, enabling the connection of sensors and the execution of automated tasks.
The code is uploaded directly to the board via USB.

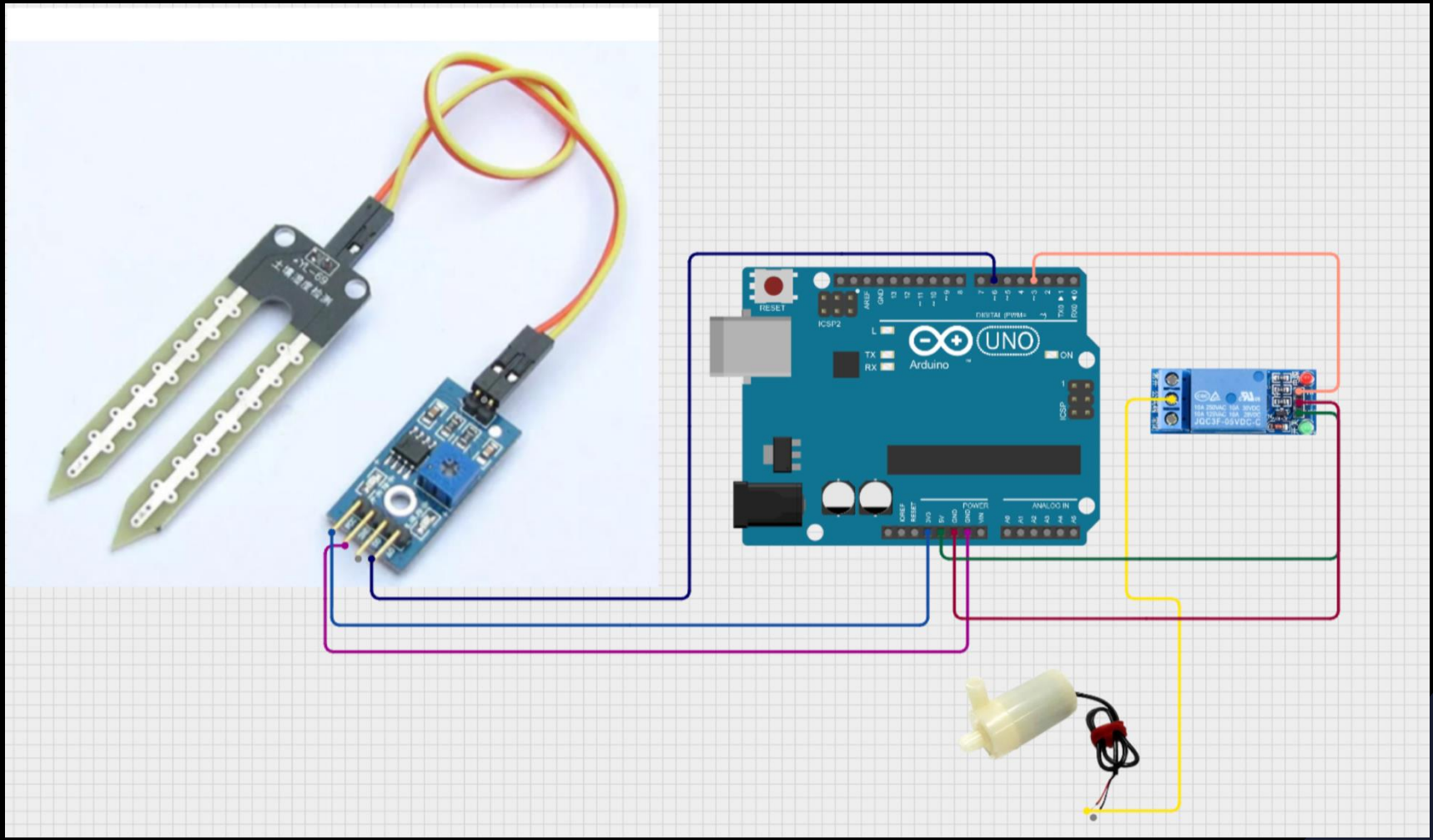
A sensor that measures soil moisture through electrical impedance.
It sends analog signals to the Arduino to determine whether the plants need watering.
It helps prevent overwatering or underwatering.




A relay that allows the Arduino to control higher-current devices such as pumps.
It acts as an interface between the Arduino’s low-voltage signals and high-voltage devices.
It ensures safe and precise on/off switching.

A DC pump that delivers water to the plants.
It is activated via the relay module when the sensor detects the need for irrigation.
It ensures efficient and consistent water supply across the plantation.

# Schematic Diagram



**Step 1:** Following the schematic shown on the previous slide, connect all your components.
**Step 2:** Upload the code to your Arduino.

# Code Explanation

**Defining Pins and Variables**
The soil sensor is connected to pin 6, and the relay module to pin 3.
The pumping duration is set to 2 seconds, and variables track the pump's status and activation time.

**Setup Function**
Serial communication is initialized, and the pins are set as input (sensor) and output (relay).
The pump starts off (Active LOW logic).

**Loop Function**
Each cycle reads the signal from the soil sensor to determine whether the soil is wet or dry.
If the soil is dry and the pump is not running, the pump is activated via the relay module.
After 2 seconds, the pump automatically turns off.

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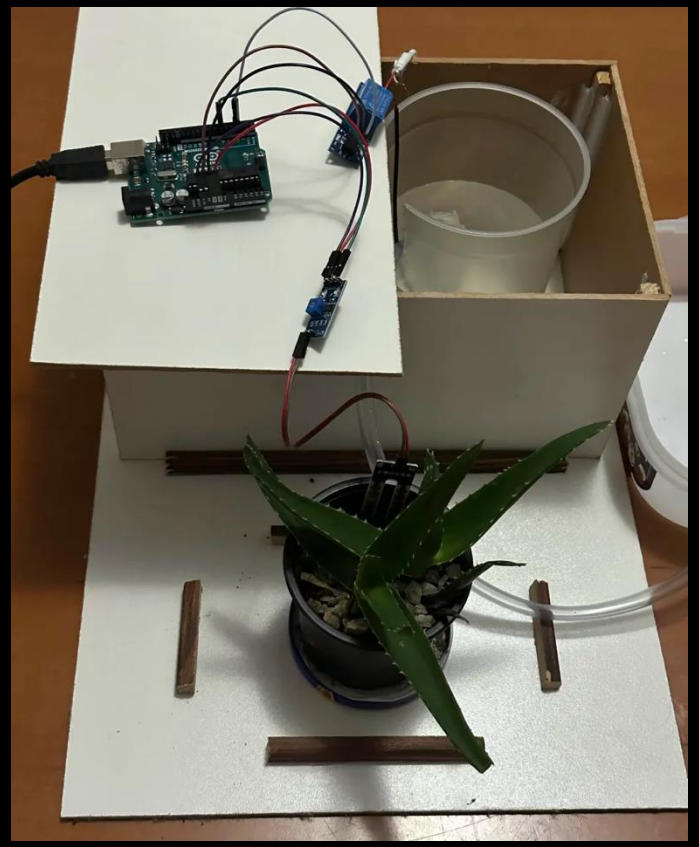
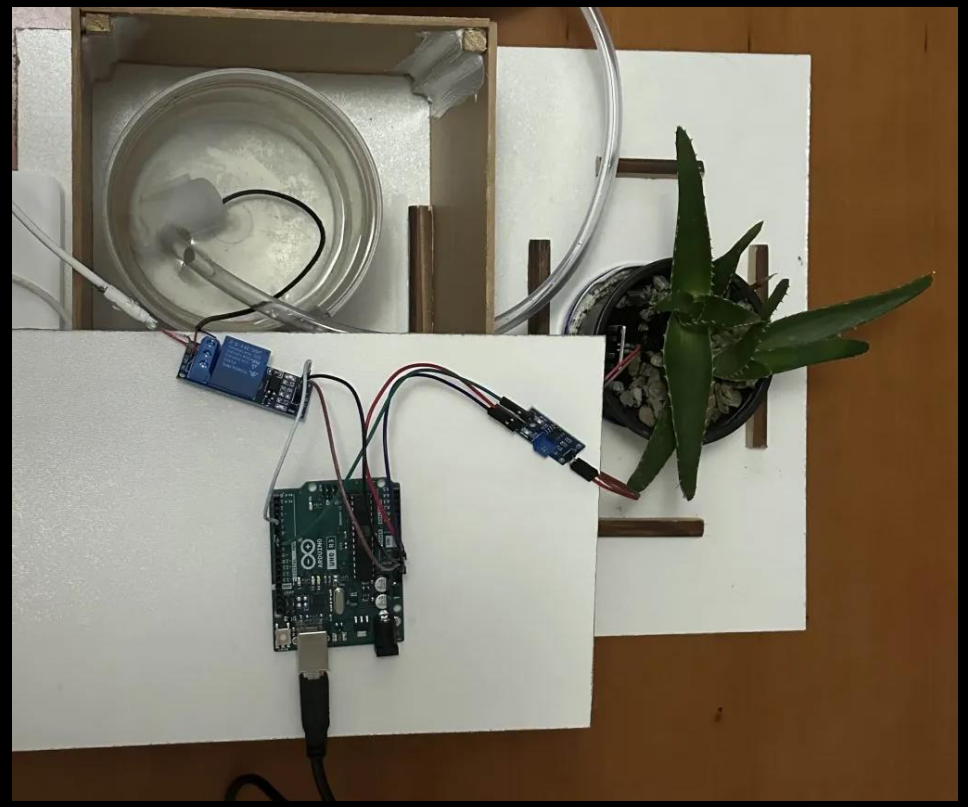
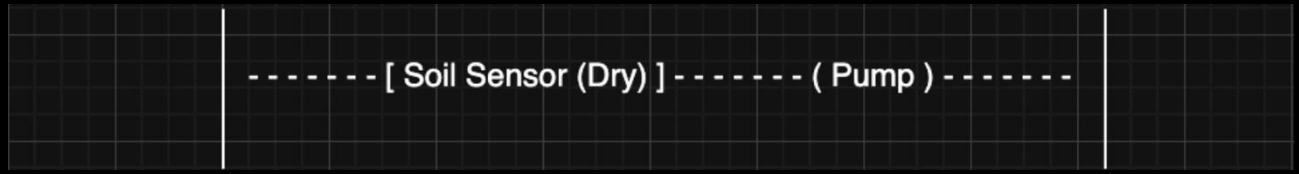
The code is modular and can be easily adapted for additional sensors and different irrigation durations.

# Final Project Layout





A ladder diagram is a graphical representation of automation logic, similar to electrical schematics. It makes the code and processes easier to understand and verify, especially for industrial systems. In my project, the ladder diagram shows how the soil sensor, relay module, and pump are connected and how they are automatically activated, enabling efficient and safe irrigation.




The ladder diagram illustrates the basic logic, which is scalable for adding new conditions and devices.